



Automated timetable generation

SKUSHINGA PHIKELELA POLYTECHNIC



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SUPERVISOR: Mrs Manzinde

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Automated timetable generation

Software Project submitted in Partial Fulfilment of National Diploma in Information Technology.

APPROVAL

I Tsitsi Priscilla Sawunyama declare that this work is original, and is not submitted to any other College for the award of any diploma.

Student signature.....Date.....

APPROVED FOR EXAMINATION

Supervisor Mrs Manzinde

Signature.....Date.....

External examiner.....

Signature.....Date.....



DECLARATION

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Chapter 1

1.1 Introduction

The Automated Timetable Generation System (ATGS) is designed to streamline and automate the creation of class schedules at Kushinga Phikelela Polytechnic. This project addresses the challenges of manual timetable creation, such as time consumption, inefficiencies, and errors by developing a software solution that uses advanced algorithms. This system will incorporate resources like rooms, equipment, and personnel, applying rules and constraints to generate accurate, conflict-free timetables efficiently.

1.2 Background:

Kushinga Phikelela Polytechnic as an institution is traditionally relying on manual processes to generate timetables, which are labor-intensive and prone to errors. Automated timetable systems, developed since the early 1960s, offer a more efficient and accurate alternative says *Ibrahim et al*

Currently, students at Kushinga Phikelela Polytechnic are almost finishing the whole semester without an official timetable and by the time they attain the official timetable, they will no longer be using it because they will already be preparing for final HEXCO examinations. Lecturers are having clashing timetables where they would have two or more lectures at the same time for example a lecturer would have a lecture to NC students and to ND students at the same time.

According to (Alkan, A., & Ozcan, E. (2003)). Manually creating an exam timetable can be slow for example taking time to add a list of all subjects and courses. Its time consuming to list invigilators (teachers) and their availability. Two exams are schedule in the same lab at the same time. Lecturer are assigned to invigilate more than one exam at the same time. Administrators are finding it difficult to adjust when last minute changes occurs such as a lecturer becoming unavailable or needing to add an extra exam. Exams for different subject are overlap especially for students who might be taking multiple subjects.

The manual timetable creation process at Kushinga Phikelela Polytechnic has become increasingly complex as the institution grows. Departments, courses, lecturers, and classroom availability all need to be considered when creating the timetable. However, this process is currently done manually, which often leads to scheduling conflicts, inefficient use of resources, and dissatisfaction among students and staff.



1.3 Problem statement:

The current manual timetable creation process at Kushinga Phikelela Polytechnic is inefficient, time consuming, and prone to errors. There are also

- Scheduling conflicts: Lecturers being double-booked, classrooms being overbooked, or courses clashing with each other.
- Difficulty in making last-minute changes: When a lecturer is unavailable or there is a sudden change in the course schedule, it becomes difficult to make adjustments without affecting the entire timetable.

1.4 Objectives

- Automate timetable generation.
- Identify and resolve clashes on timetable (ensure there are no clashes on timetable).

1.5 Research Propositions / Hypotheses

The research hypotheses guiding this project are as follows:

1. H0 (Alternative Hypothesis): Automating the timetable creation process significantly reduces scheduling conflicts and improves resource utilization at Kushinga Phikelela Polytechnic.
2. H1(Null Hypothesis) : Automating the timetable creation process does not significantly reduce scheduling conflicts or improve or resource utilization at Kushinga Phikelela Polytechnic

1.6 Justification of the Research

The research is justified on several fronts:

- Efficiency: Manual timetable creation is inefficient and prone to errors. Automating this process will save time and reduce the workload on administrative staff.
- Error Reduction: By automating the process, the potential for human errors.
- Scalability: As the institution grows, the complexity of creating timetables increases. An automated system can scale more easily to accommodate this growth.



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- Improved Resource Utilization: The system will ensure that resources such as classrooms and lecturers are optimally allocated, reducing downtime and improving overall efficiency.
- Potential impact on educational outcomes: By reducing scheduling conflicts and ensuring that tutors are correctly assigned, the system can contribute to better educational outcomes for students.

1.7 Delimitation

The automated timetable generation system will help create schedules for all departments such as IT and Horticulture .The system will generate timetable ,identify and solve clashes .

1.8 Assumptions

The following assumptions have been made for the development of the timetable generation system:

- All lecturers and classrooms will be available. as per the data supplied by the administrators
- Courses and their respective time requirements are well-defined before the generation of the timetable.
- Sudden changes in lecturer availability will be communicated promptly so that adjustments can be made in real-time.
- The institution has the necessary infrastructure (computers, network, etc.) to support the automated timetable system.

1.9 limitations of the Study

One of the challenges encountered was the lack of consistent internet access , which restricted the ability to gather online resources .Additionally , frequent power outages disrupted the workflow , delaying the development and testing of the Automated Timetable Generation.

1.10 Conclusion

The chapter has outlined the key components of the research project, including the introduction of the problem, research objectives, and justifications for conducting the study. The next chapter will provide a detailed review of the literature related to timetable automation systems and relevant scheduling algorithms.



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Chapter 2:

1.11 Feasibility study

The proposed Automated Timetable Generation System (ATGS) aims to automate the creation of class schedules at Kushinga Phikelela Polytechnic. This feasibility study will assess the system's work-ability, its alignment with user needs, effective use of resources, and overall cost-effectiveness. According to (*Narang, Gupta and Bansal 2012*) feasibility study is categorized into technical, operational, economic, schedule, and social aspects to ensure the proposed system meets all requirements and delivers value.

1.11.1 Technical Feasibility

Objective: Determine if the problem can be solved using the existing technology and resources available.

Table 2-1: Elements required to develop the system

Computer elements	Qty	Availability	Commentary
Hardware requested			
HP color LaserJet Ent CP5525dn	4	Yes	Additional 4 Laser-jet CP5525 may be needed
HP Pro One 600 All in one	8	Yes	
Cisco 2900 series router	1	Yes	
PS2 KVM Console switch	4	No	Organization can procure extra PS2 KVM Console switch

Technical Expertise: The institution will need personnel proficient in developing and maintaining optimization-based software, although this can be outsourced if necessary

- Conclusion: The system is technically feasible given the existing technology and resources available at Kushinga Phikelela Polytechnic.

1.11.2 Economic Feasibility

Objective: Evaluate whether the likely benefits outweigh the costs of developing the ATGS system, generally through a cost-benefit analysis.



1.11.3 Operational Feasibility

Objective: The researcher carried out the study to analyze and determine whether the needs of the Kushinga can be met by completing the project. A user-friendly system is required so that it can be quickly accepted.

- **User Requirements:** The system directly addresses user concerns such as scheduling conflict and time consumption by implementing real-time adjustments and conflict-free timetables, the system meets the operational needs of administrators, lecturers, and students.

1.11.4 Cost benefit analysis (CBA)

It is a technique for assessing the monetary social costs and benefits of a capital investment project over a given time period. A decision to proceed with a project can only be made where the benefits outweigh the costs.



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Cost analysis table for outsourcing

Table 2-2: Cost Analysis for outsourcing

	Year 1 USD	Year 2 USD	Year 3 USD
Reduction in stationary costs	1 000	950	950
Estimated saved expenses	700	500	450
TOTAL BENEFITS	1 700	1 450	1 400
DEVELOPMENT COSTS			
System development	200	-	-
Training	250	-	-
Additional software	600	300	300
TOTAL DEVELOPMENTAL COSTS	1 050	300	300
OPERATIONAL COSTS			
Hardware maintenance		200	150
Software maintenance		100	90
Other maintenance		90	50
Internet Service Provider(Telone)	700	500	300
Stationary and computer consumables	150	100	100
TOTAL OPERATIONAL COSTS	850	990	690
TOTAL COSTS	1 900	1 290	990
[BENEFITS] less [EXPENSES]	(200)	160	410

1.11.5 Return on Investment (ROI)

Return on investment refers to the profitability of the project expressed in percentage. Return on investment is used to calculate the viability of the project.

$$\text{Return on Investment} = \frac{\text{Net income}}{\text{Investment Cost}} * 100 \%$$

Investment Cost

$$\text{Return on Investment} = \frac{(200)+160+410}{4780} * 100 \%$$

$$= 7,74\%$$



The ratio of income generated to investment is sufficient. The project is economically feasible since the return on investment is 7,74%

1.11.6 Payback period

This refers to the period it takes the cash inflows from a capital investment to equal the cash outflows, usually expressed in years. It is wise to accept the project with the shortest payback period. Payback Period (in years) = $\frac{200}{12}$

1450

Payback Period (in years) = 1 year 6 months

Net present value

Table 2-3: Net present value

Year	Cash flow
0	-1900
1	1700
2	1450
3	1400

$$\text{Present value} = \frac{\text{future value}}{(1+i)^n}$$

$$\text{Present value 0} = -1\,900$$

$$\text{Present value 1} = 1700 / (1+0.2)^1$$

$$= 1\,416,66$$

$$\text{Present value 2} = 1450 / (1+0.2)^2$$

$$= 1\,006,94$$

$$\text{Present value 3} = 1400 / (1+0.2)^3$$

$$= 810,19$$



Net present value

Table 2-4: Net present value

Year	Cash flow	Present value
0	-1 900	-1 900
1	1 700	1 416, 66
2	1 450	1 006, 94
3	1 400	810, 19
NPV		321, 479

1.12 Schedule Feasibility

The researcher calculated and continually re-examined whether it is possible to complete all amount and scope of work lying ahead, utilizing the given number of resources, within required period of time.

- **Development Time:** Developing an automated timetable system using genetic algorithms and constraint-based models typically takes 6-12 months, depending on the complexity of Kushinga Phikelela scheduling needs.
- **Implementation Time:** Once developed, implementation across the institution (including testing, staff training, and data migration) should take an additional 3 months. A phased roll-out is recommended to minimize disruption.
- **Risk Management:** Delays could arise from technical challenges, such as integrating real-time updates or handling last-minute scheduling changes. Proper project management and risk mitigation strategies will be necessary.
- **Conclusion:** The system is schedule feasible, with a realistic timeline for development and implementation. Any minor delays can be managed with proper planning.

1.12.1 Work plan

Is a work plan shows all the activities to be taken from the start up to the end of the chronological order



Table 2-5: Work Plan

	Phase name	Phase activities	Activity duration days	Phase duration
1	Proposal	Background	17	July
		Problem definition	8	
		Objectives	15	
2	Feasibility study	Technical Feasibility study	4	August
		Economic feasibility study	3	
		Operational feasibility	8	
		Schedule Feasibility study	10	
		Project plan drafting	5	
3	Requirement and analysis	Data gathering	13	September
		Process analysis and Modeling	9	
		Data analysis and modeling	8	
4	Design phase	Process modeling for proposed system	15	October
		Data modelling for proposed system	10	
		Hardware and software specification	12	November
		Architecture design	15	
		Program specification and design DFD to second level	17	December
		Data store and Interface design and UML diagram Output diagram	14	
5	Coding and testing	Staff and user training	10	January
6	Implementation Evaluation and maintenance	Evaluations Audits and Maintenance	30	March



1.12.2 Gantt chart table

Table 2-6: Gantt chart table

Time period in									
Months	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Marc
Proposal									
Feasibility study									
Requirement analysis									
Design									
Coding and testing									
Implementation and post implementation plan									
Summary , conclusion and Recommendations									

1.12.3 Social Feasibility

Objective: Determine if the system can be introduced without causing social issues and whether it will be acceptable to the institution and broader society.

- **User Acceptance:** The system will likely be well-received by both staff and students. Administrative staff will benefit from reduced workload and increased accuracy, while students will appreciate having well-organized timetables.
- **Impact on Work Culture:** The system may require administrative staff to adjust to new processes. However, since it reduces the burden of manual scheduling, staff are likely to adapt positively.



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- ***Social Concerns:*** There are no significant social concerns associated with the implementation of this system. Its introduction will enhance the institution's reputation for using advanced technology to improve operational efficiency.
- ***Conclusion:*** The system is socially feasible. It addresses key issues without causing any significant social concerns and will likely be embraced by the users.



1.13 Requirement analysis

1.13.1 Introduction

Petrovic et al (2012) highlighted that ,what users required in the proposed system and gave an insight of the operations of the current system, how processes are linked within the current system and how activities are going to be coordinated in the proposed system. The researcher used a set of gathering methodologies to collect data. The researcher analyses collected data highlighted the advantages and disadvantages of the current system and alternative solutions to the systems weaknesses.

1.14 INFORMATION GATHERING METHODOLOGIES

1.14.1 Interviews

On (1 August 2024 up to 7 August 2024) the researcher asked questions to twenty staff members regarding their opinion on the efficiency and effectiveness of the current system and their desired changes. Interviews were carried in a space of one week, and all the staff members were interviewed in their offices. Notes were taken down during all interview sessions. The developer chose to interview the IT unit department and administrative staff members separately so as to enable the gathering of all information from different levels and sources related to the manual timetable , operations and activities undertaken by the institution. A variety of questions were open-ended questions, closed-ended questions, and range of response questions. Interviews were documented and recorded during and after the process of interviewing.

Benefits

- Managed to clarify misconceptions about the current system

1.14.2 Questionnaires

Researcher gave questionnaires to all staff members at Kushinga Phikelela. The purpose is to preview into the already existing system through the users' opinion. Questionnaires were to be returned in five days' time.

1.14.3 OBSERVATION

The researcher simply observed all activities that takes place at Kushinga Phikelela and managed to have the look and feel of what was actually transpiring .



1.15 ANALYSIS OF THE EXISTING SYSTEM

Kushinga Phikelela currently used an inventory management system as their system to manage and keep records pertaining their products.

1.15.1 DESCRIPTION OF THE EXISTING SYSTEM

The researcher carried out the analysis to depict how the current system operates and explores the weaknesses of the current system so that modalities, opportunities and improvements can be done on some modules of the current system.

1.15.1 ANALYSIS OF THE CURRENT MANUAL TIMETABLE

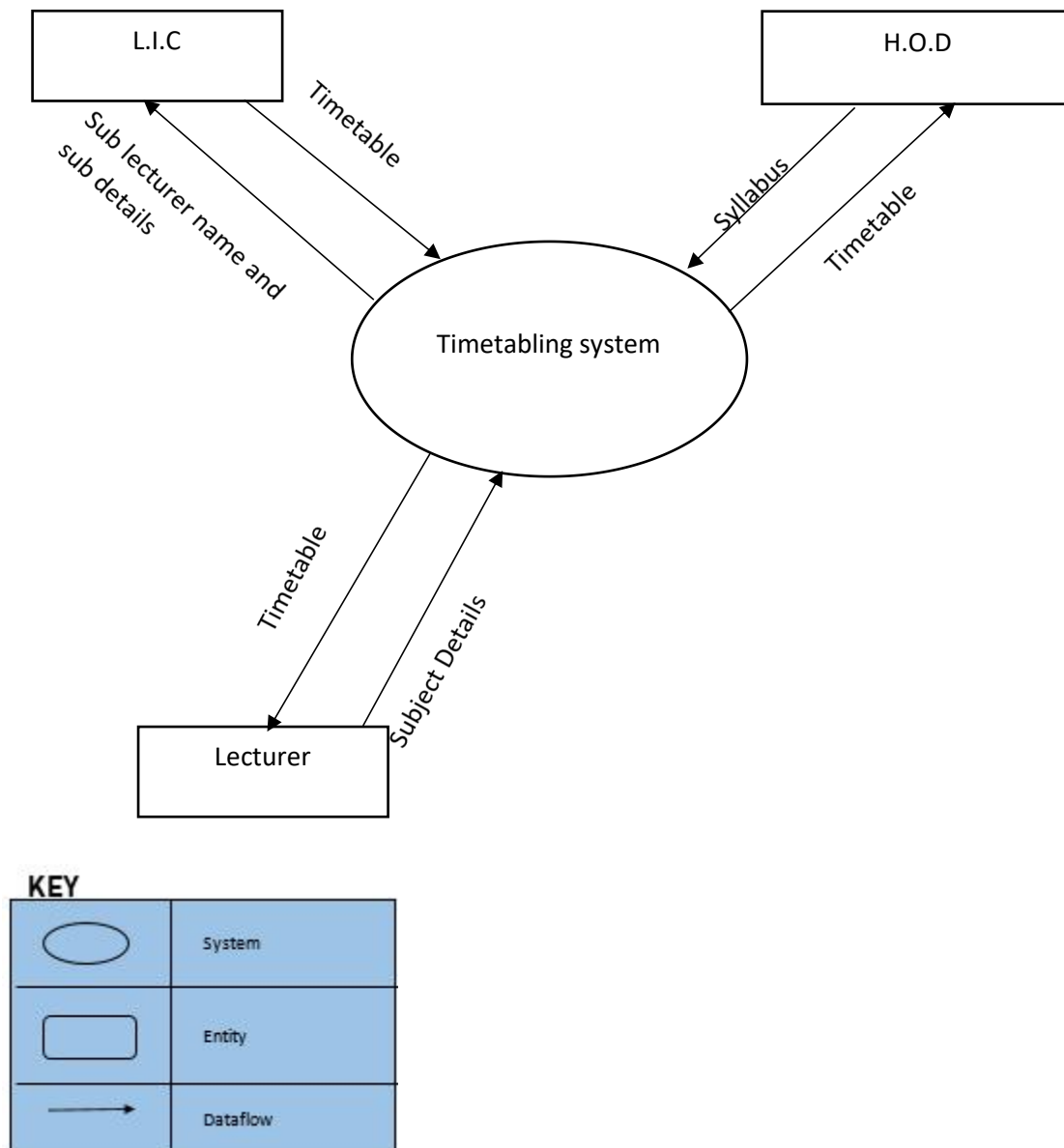


Figure 3:1: Context diagram



FUNCTIONS

L I C

- Someone who create a timetable
- Receive lecturer's details and lecturer's subjects
- Update ,delete and modify the timetable

Lecturer

- Receives timetable
- Send his/her details to the LIC
- Send subjects

HOD

- Send the syllabus to L I C
- Receive the timetable



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Use case diagram

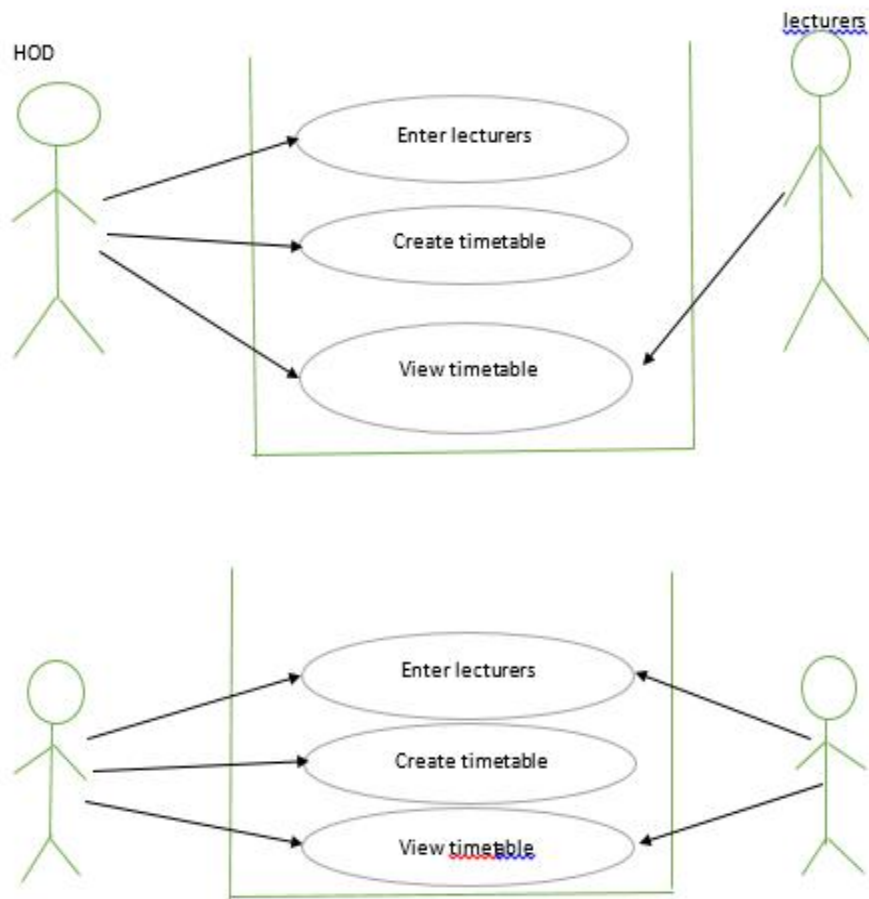


Figure 3:1: Use case diagram 1



Use case diagram

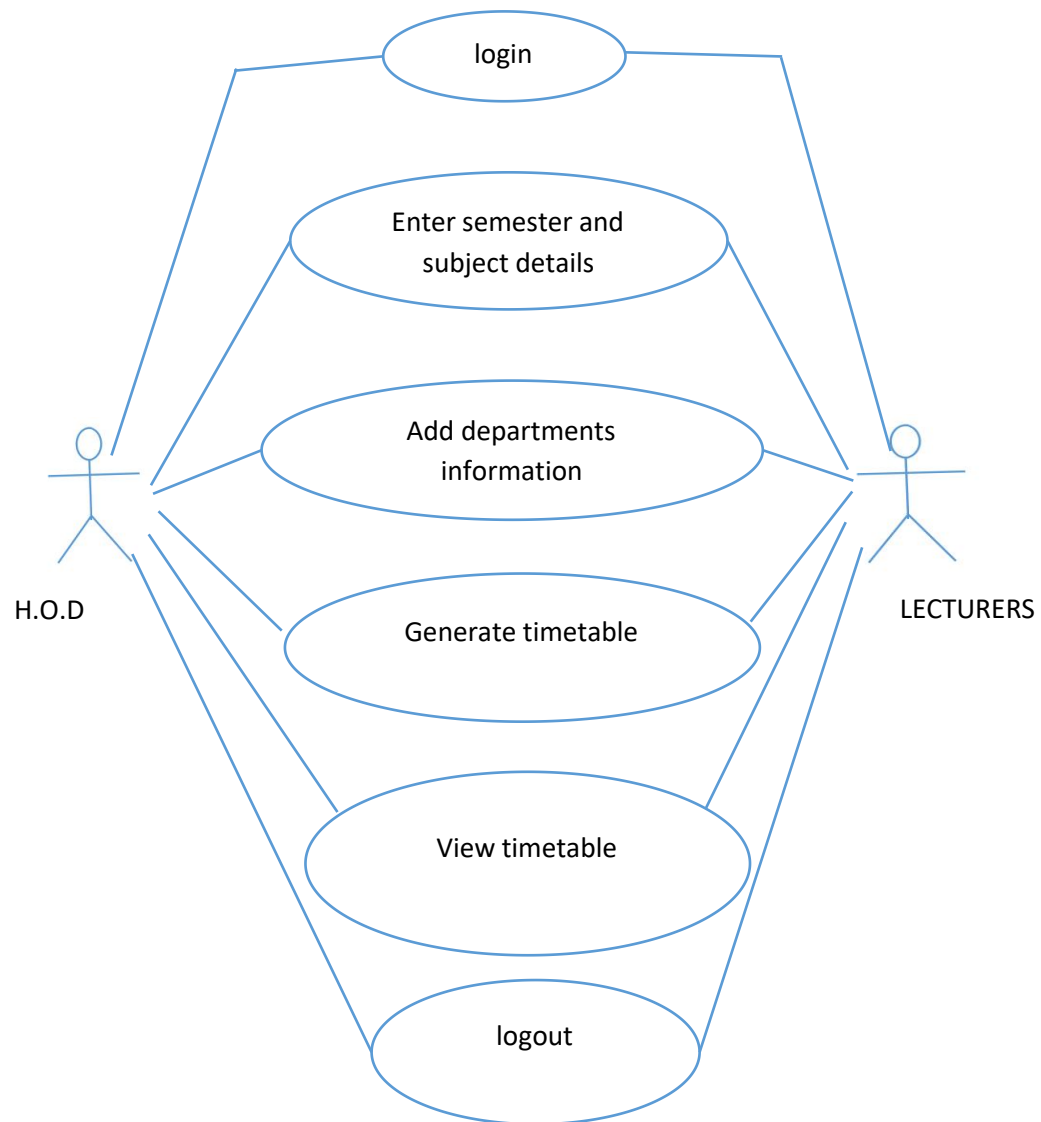


Figure 3:2:: Use case diagram 1

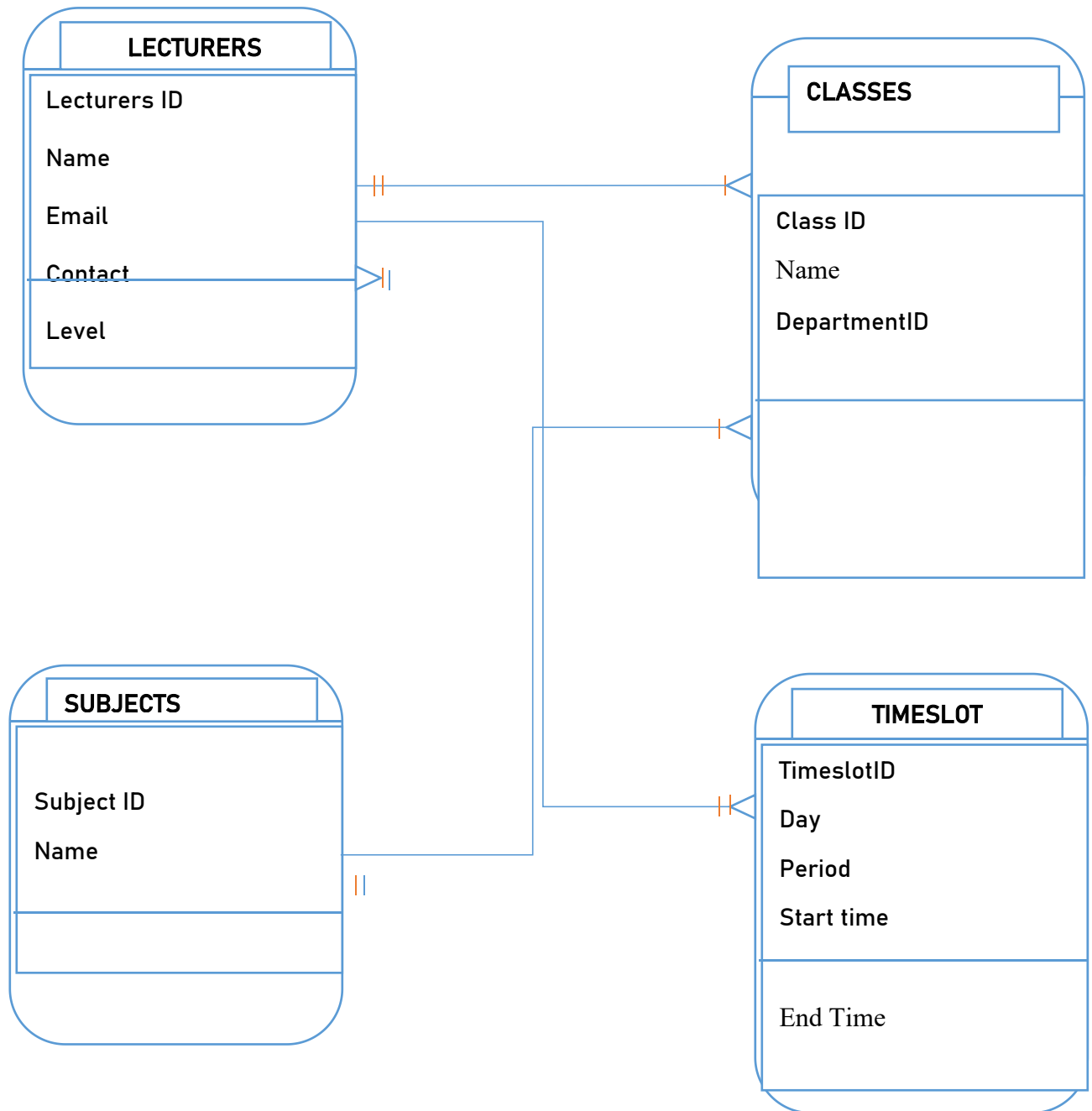


Figure 3:3:Class diagram : source <https://scispace.com>

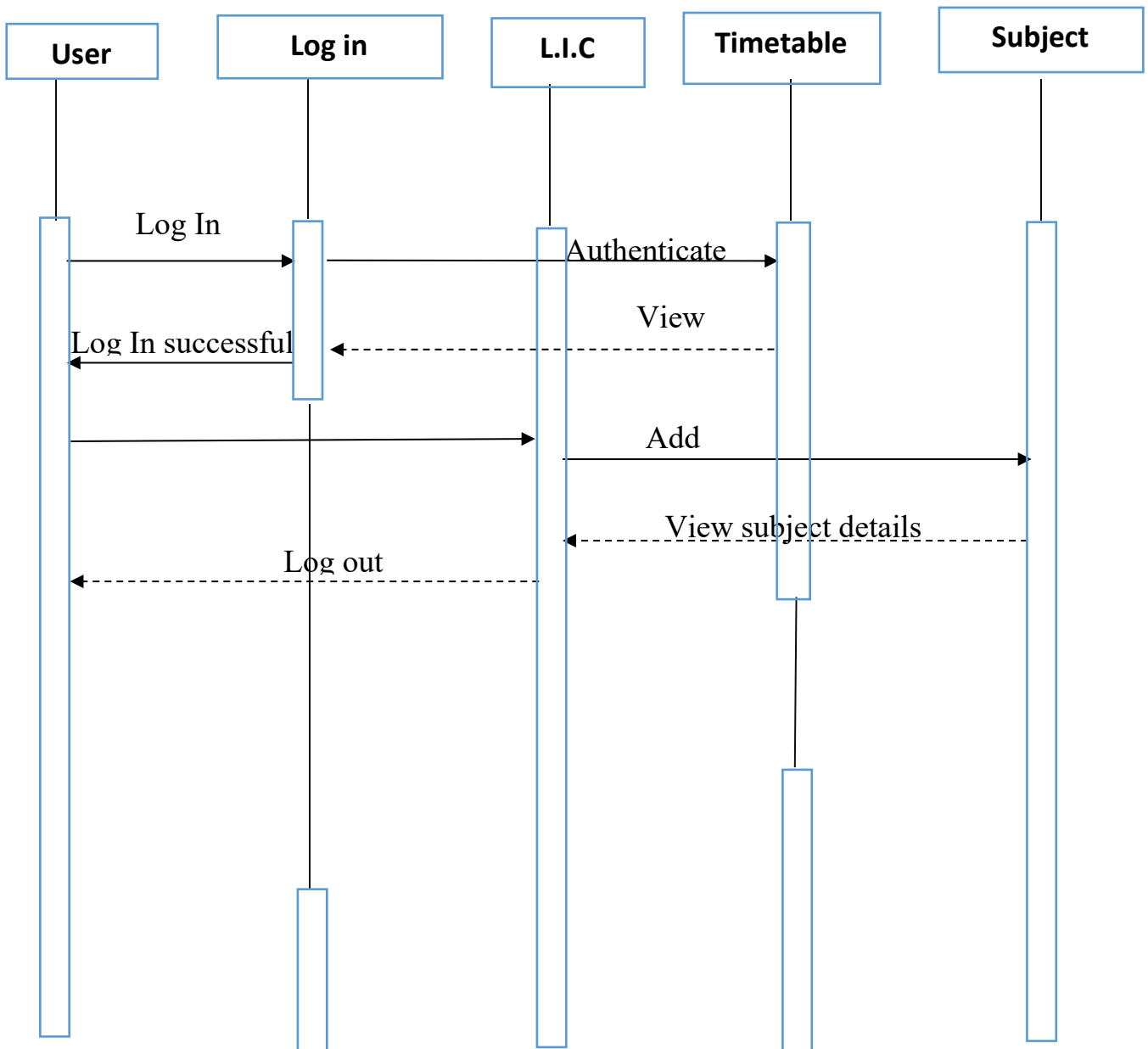


Figure 3:4: Sequence diagram: source <https://www.irjmets.com>



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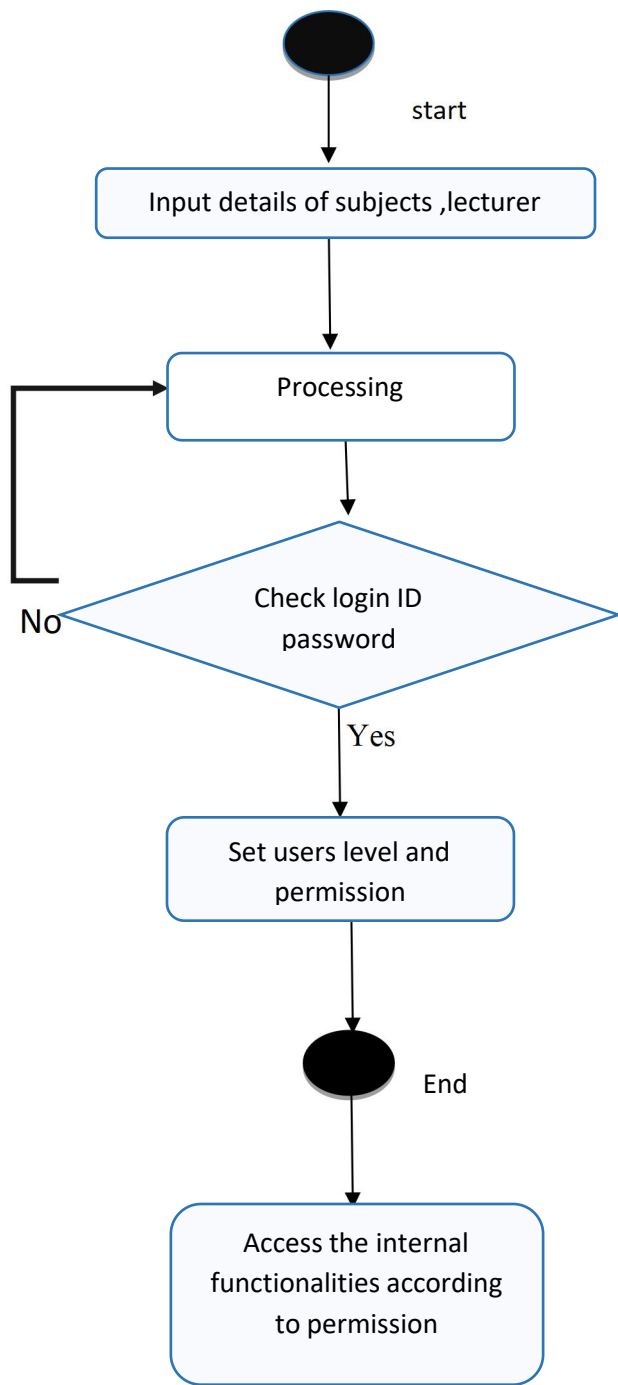


Figure 3:5: Activity diagram



1.15.2 EVALUATION OF ALTERNATIVES

The developer considered the following alternatives to the development of the proposed system:

Outsourcing Improving the current system In house development

1.15.2.1 Outsourcing A.s.c Software

The developer assumed to buy a software that is ready-made product.

Benefits

- No development costs incurred.
- Less technical staff is required.
- System improvements can be done on user request.

Drawback

- Outsourcing may result in increased risk of information leakage and reduce confidentiality
- Results in more cost being incurred because of high maintenance

1.15.2.2 In-house development

The developer came up with a system using company's internal resources and pool of IT resources.

Benefits

- System users assumed complete ownership of the system as system specifications are based on user requirements during system development.
- Kushinga Phikelela would continue to build its employee skill base through in house development and training of staff.
- Implementation would be easier since the personnel responsible for the system development will be readily available.

Drawbacks

- Resistance from employees might be faced
- Time consuming that is development of the system would take time as compared to other alternatives.



SYSTEM DEVELOPMENT CHOICE

The choice of development would be in-house because of the following reasons:

Cost- It is very cost effective to develop an own internal system as the costs are shown in the table below:

Table 3-1: Alternative Table

Alternative	Cost \$
Improvement of the current system	1500
Outsourcing ASC	3000
In house development	800

Budget

In house development would be the best alternative because it falls within the budget given by the current budgetary constraints.

Expertise

All the needed technical expertise in personnel like programmers and analysts are available at the College.

1.16 REQUIREMENT ANALYSIS FOR THE PROPOSED SYSTEM

Requirement analysis can be categorized into two phases namely:

Functional requirements

The researcher captured the intended behavior of the system. The system should offer the following services and functions:

- . The ability to store data to the system.
- . Use of passwords and usernames to access
- . Allow compilation, updating and retrieval data.
- . To provide analytical inventory information.

Non-functional requirement



The system should have the following features:

- The implementation should support the Microsoft Windows family of operating systems with backward compatibility.
- User friendly interface
- Error handling capabilities
- The system should have access levels.

USER REQUIREMENTS

In order to be able to use the system the user has to:

- Be able to enter correct information in the system.
- Be able to retrieve information from the database
- Be able to update the database on each and every timetable.

1.17 CONCLUSION

Important data has been collected using various data gathering methodologies and that data played a pivotal role in the analysis of the old system using various diagrammatic and logical methods. The evaluation of alternatives has also revealed the need of the in-house development of the new system against the popular and attractive options of outsourcing and upgrading the existing system hence endorsing the proposed plan.



1.18 Design

Introduction

The researcher focused on how the proposed system is going to work and how it is going to be constructed or designed.

System Inputs, Processes, Outputs, and Interfaces

- **Inputs:**

The developer designed the system in such a way that it should input data before processing that is login. Data will be validated before being accepted by the system to ensure that correct and standardized data is stored within the database.

LOGIN WINDOW

ATGS

KPP TIME TABLE GEN SYSTEM(LOGIN)

USERNAME

PASSWORD

LOGIN

Figure 4:1: Login page



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-Users should be prompted to enter usernames and password, access level will determined using the provided credentials. Giving incorrect username and password would result in access being denied.

- Data about lecturers, classes and courses, including time slots and availability.
- Real-time updates for last-minute changes (e.g., lecturer unavailability).

1.18.1 Login Code

See APPENDIX D

The developer designed the system in a way that it carries many processes.

1.18.2 Pseudo code

Connection to the database

Check for database connection establishment

If they is no connection yet

Set the connection

Else

Report an error

User login

Enter identification information (username and password)

If correct proceed to the next module

Else

Report error then try again



User new Entry

Enter required information

If invalid data is required

Report error

Else

Save data (using the established database connection)

- Schedule generation using genetic algorithms and constraint-based models.
- Conflict resolution for overlapping classes, resource optimization, and last-minute adjustments.

- **Outputs:**

After input and processes the system output information to users in various output designs according to the action being carried out.

1.19 Database architecture design

Efficiency and fastness in record saving and retrieving or manipulation starts on the design of a good database and it is only a well-designed database that will avoid system crashes and updating anomalies.



1.19.1 Physical design

The developer is concerned with how hardware components are going to communicate.

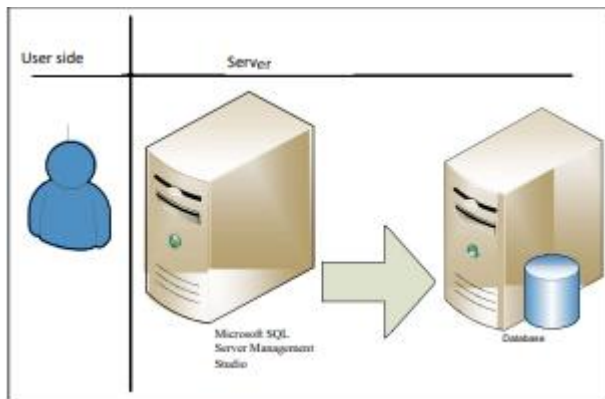


Figure 4:2: Physical design. Source <https://scispace.com> page 53



Database architecture design

The developer used the three-level schema architecture to describe the database specified during designing the database. The diagram below shows the database architecture of the proposed system. Source <https://www.irjmets.com>

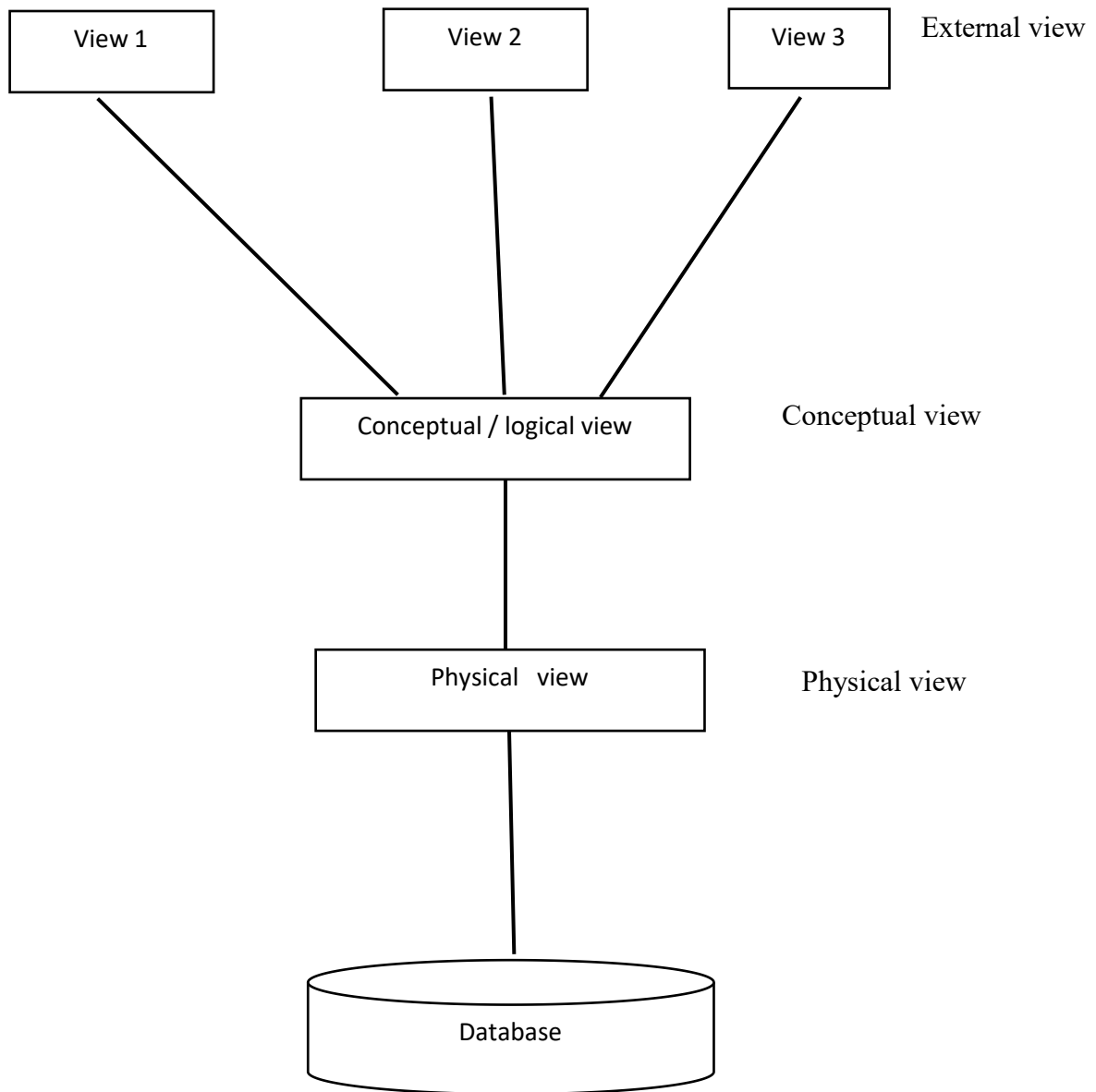


Figure 4:3: Database architecture design



Physical level

Lowest level of abstraction and defines how data is actually stored.

Conceptual level

Described what data is actually stored in the database and the relationships that exist among the data.

External level

Database records can be viewed using forms and reports. The database manipulation is done using Structured Query Language (SQL)

1.19.2 DATA DICTIONARY

Table 4-1: Data dictionary

Class ID	Name	DepartmentID
NC1.1	FEBRUARY	ATGS0001DEP
NC1.2	FEBRUARY	ATGS0001DEP
ND2.1	FEBRUARY	ATGS0001DEP
ND2.2	FEBRUARY	ATGS0001DEP



Table 4-2: Lecturers

LECTURER ID	NAME	EMAIL	PHONE	LEVEL
ATGS001	P Manzinde	pmanzinde@gmail.com	0772913473	HOD
ATGS002	F Maviza	fmaviza@gmail.com	0773791133	
ATGS003	S Chabanga	schabanga@gmail.com	0772924331	L.I.C
ATGS004	S. Ndlovu	hndlovhu@gmail.com	0775870578	
ATGS005	C. Huruva	dhuruva@gmail.com	0782689674	
ATGS006	N Chekai	hchekai@gmail.com	0783027434	
ATGS007	J Chigu	fchigu@hamil.com	0714537865	



Table 4-3: Class subject

CLASS ID	SUBJECT ID
NC1.1	553/01/S01
NC1.1	553/01/S02
NC1.1	553/01/S03
NC1.1	553/01/S04
NC1.2	553/02/S01
NC1.2	553/02/S02
NC1.2	553/02/S03
NC1.2	553/02/S04
ND1.2	553/03/S01
ND1.2	553/03/S02
ND1.2	553/03/S03
ND1.2	553/03/S04
ND2.2	553/04/S01
ND2.2	553/04/S02
ND2.2	553/04/S03
ND2.2	553/04/S04



Automated timetable generation

Table 4-4: Users

#	NAME	TYPE	COLLATION	NULL	DEFAULT
1	id	Varchar(20)	Utf8mb4_general_ci	No	None
2	name	Varchar(20)	Utf8mb4_general_c	No	None
3	username	Varchar(20)	Utf8mb4_general_c	No	None
4	password	Varchar(20)	Utf8mb4_general_c	No	None

Table 4-1: Time slot

#	NAME	TYPE	COLLATION	NULL	DEFAULT
1	Timeslot	Varchar(30)	Utf8mb4_general_c	No	None
2	Day	text	Utf8mb4_general_c	No	None
3	Period	Int(11)		No	None
4	Start time	time		No	None
5	End time	time		No	None



Table 4-2: Subjects

SUBJECT ID	SUBJECT NAME
553/01/S01	NASS
553/01/S02	CPP
553/01/S03	Comp system
553/01/S04	SHEQ
553/02/S01	Comp Sec
553/02/S02	ESD
553/03/S03	DBase
553/04/S04	Comp net
553/03/S01	H Admin
553/03/S02	OOP
553/03/S03	D Base
553/03/S04	D Admin
553/04/S01	Soft Eng
553/04/S02	WEB
553/04/S03	Design Analyses
553/04/S04	OS Admin



Table 4-1:User interface Design

The developer allowed users to interact with the system through the interface. The developer supported the following user interface features:

- **User friendly**

The interface provided necessary elements and clear language used on labels and in messaging.

- **Consistency**

Once users learn on how to do something, they would be able to transfer that skill to other parts of the system.

- **Attractiveness**

Attracted Interface design.

- **User guidance**

The interface provided guidance on how to use it.

- **Maintainable**

The Interface accepted changes to be integrated without causing a conflict of interest.



Processing controls

The developer designed the system which validates data entered by the user. Some processes would completely fail if wrong data is entered and it is only through validation that correct data is entered into the system. The manipulation and processing of data is to be done by authorized users only who are to do that after a valid login.

Storage controls

The developer provided the system which supported data backups and restore facilities. Those controls were used to avoid the complete data loss when the database crushes or faces an error.

Security

File Security

Files were electronically stored inside the database. Electronic database means easy and fast backup. Database stored on a server computer in the server room and the server room stay locked and accessed by only the allowed staffs.

Software security

The developer installed Anti-virus packages to protect software against viruses and other suspicious software attacks. The developer tested software before implementation to assure that the software cannot be easily hacked.

File backup

Ways to store backup:

The developer used the following types of media for backup:

- External hard disks.
- DVDs and CDs.
- Secure Network locations.



Physical security

The developer used the following for physical security:

- Used uninterrupted power supply to remove instant and complete data loss when electricity power cuts.
- Door locks and keys to lock the server room.
- Maintained favourable temperatures needed to be regulated inside the server room to avoid over heating.



1.22 ENTITY RELATIONSHIP DIAGRAM.

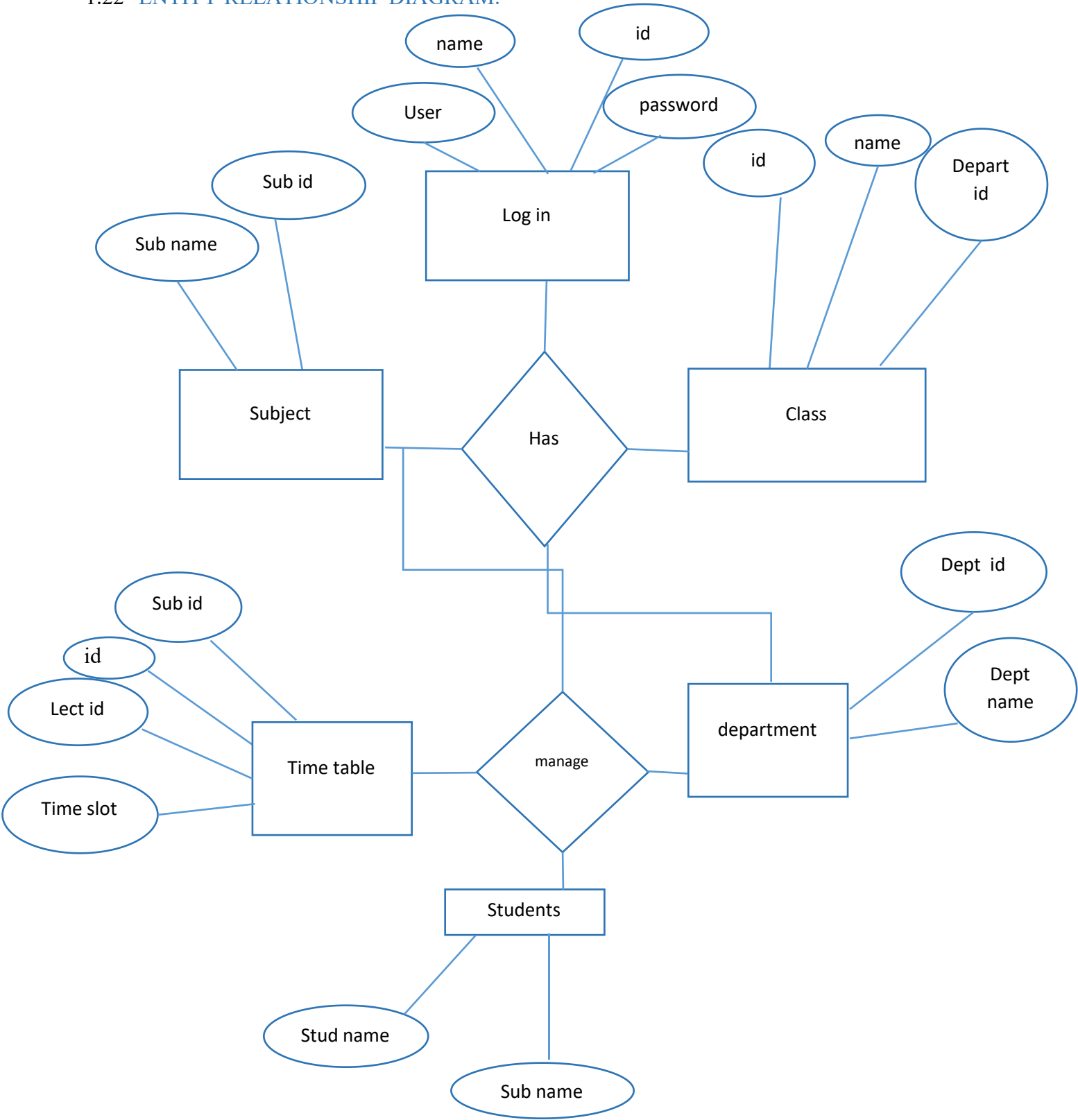


Figure 4:1: Entity Relationship



1.23 Data flow diagram

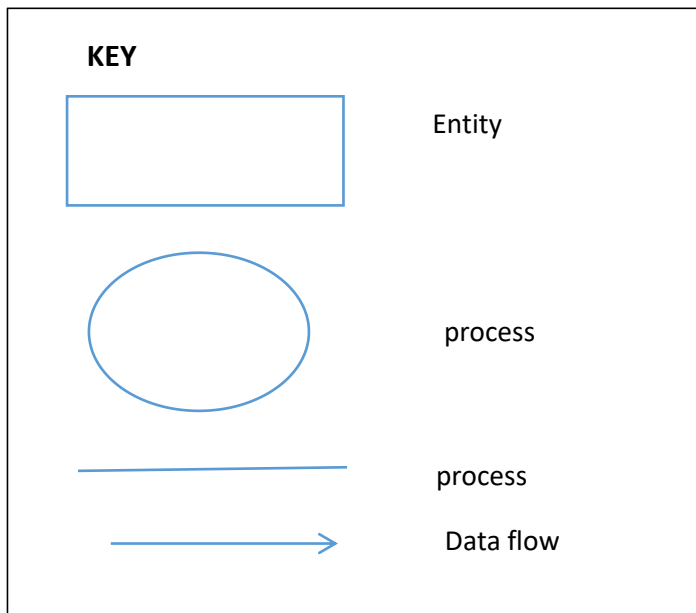
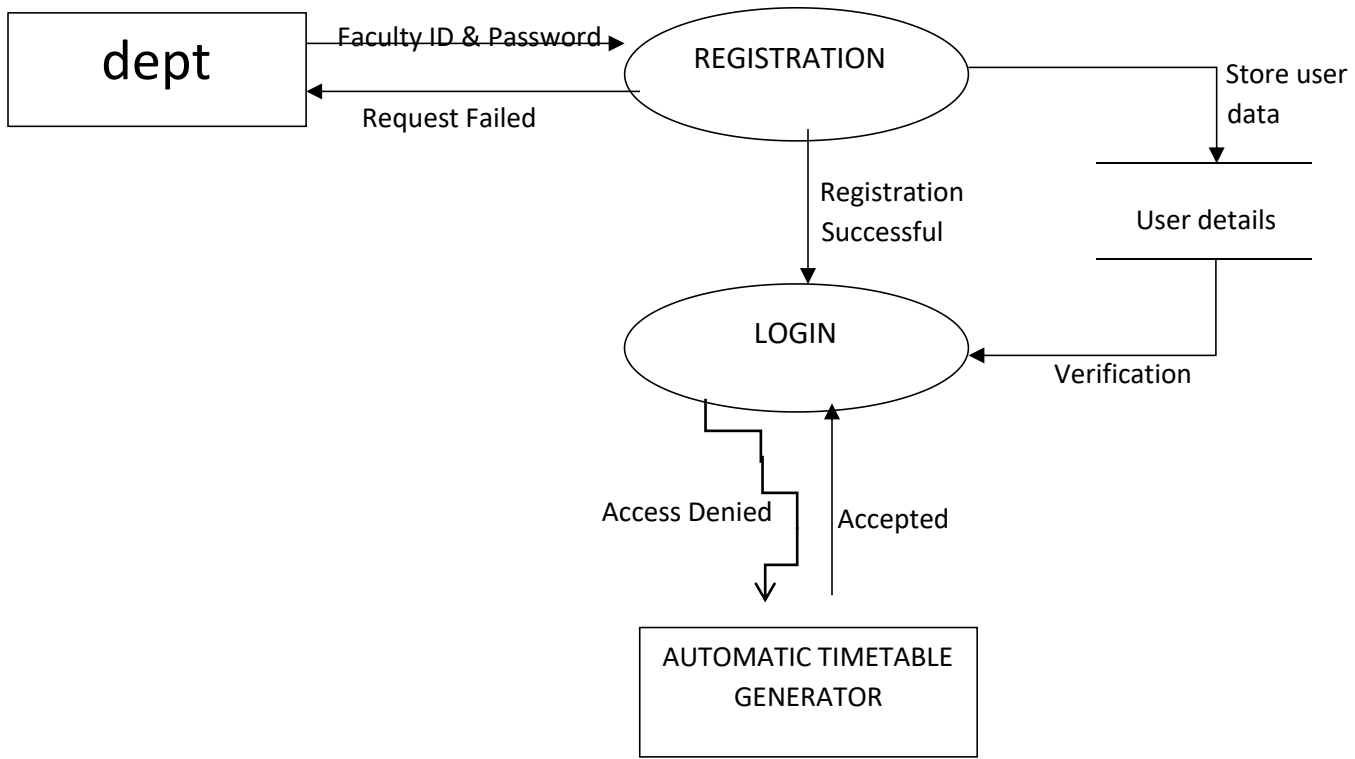


Figure 4:2: DFD LEVEL 1



DFD LEVEL 2

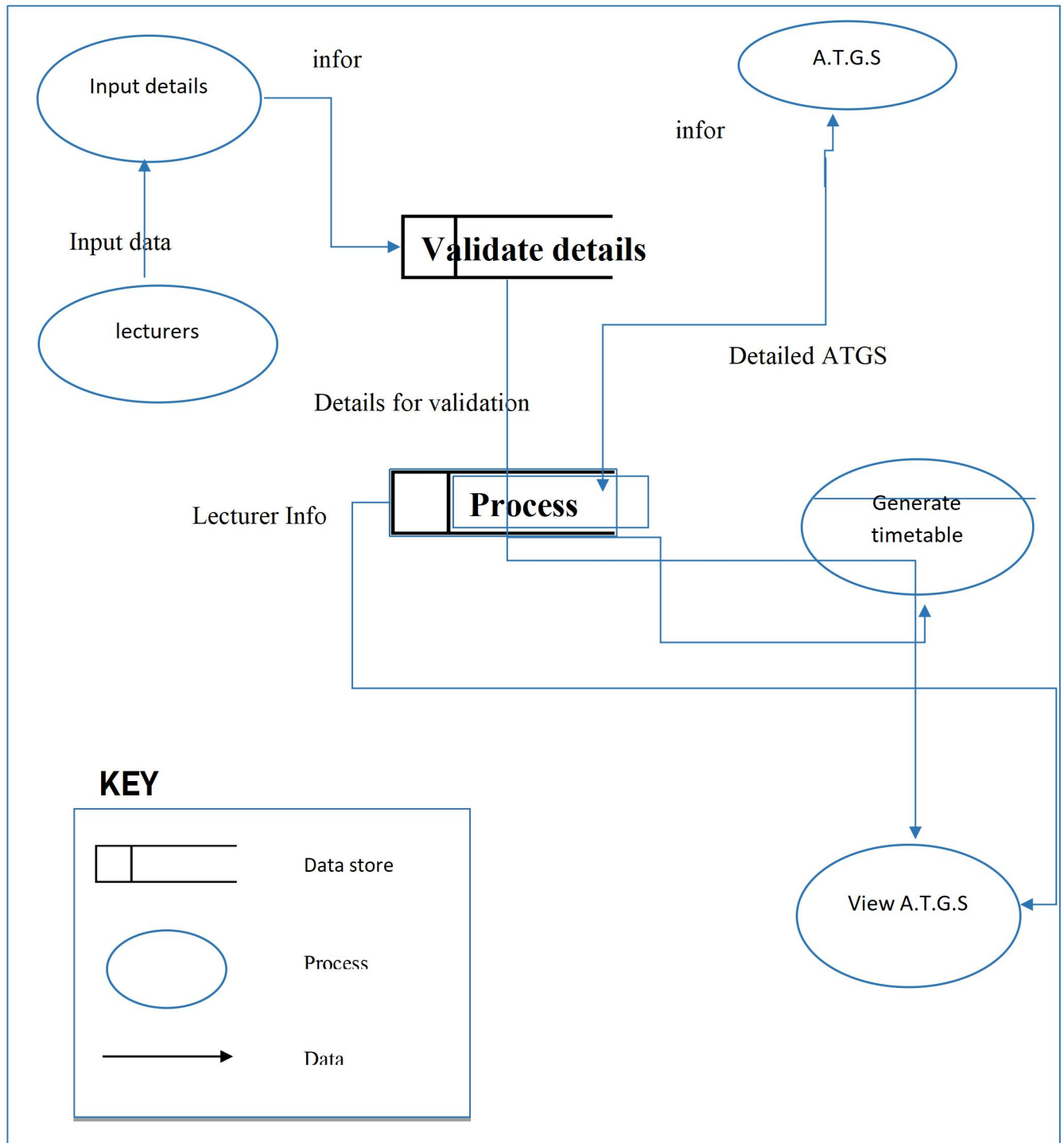


Figure 4.3: DFD LEVEL 2: source <https://scispace.com> page 87



UML diagram

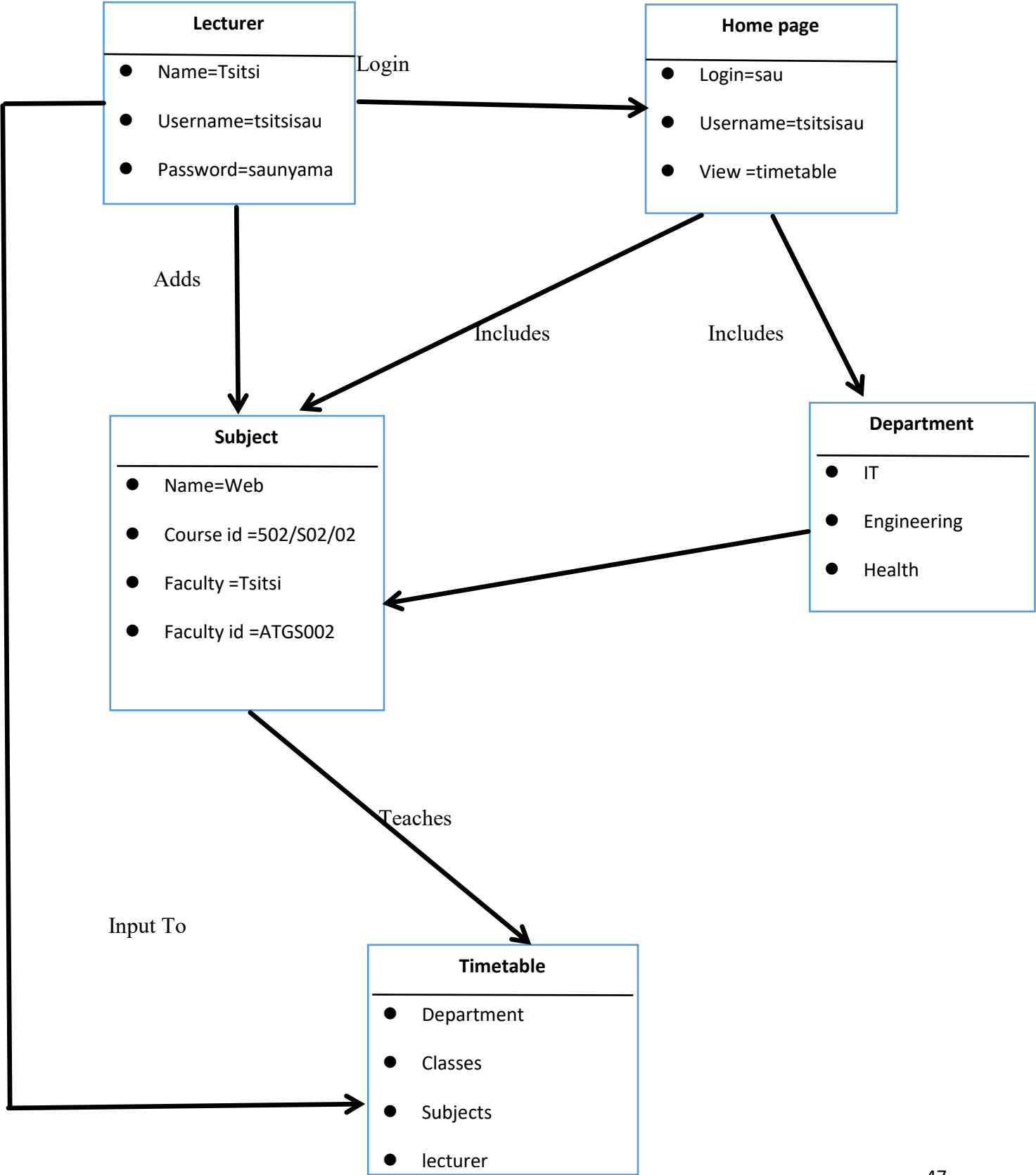




Figure 4:4: UML Diagram : says by Petrovic et al (2012)

Data flow diagram

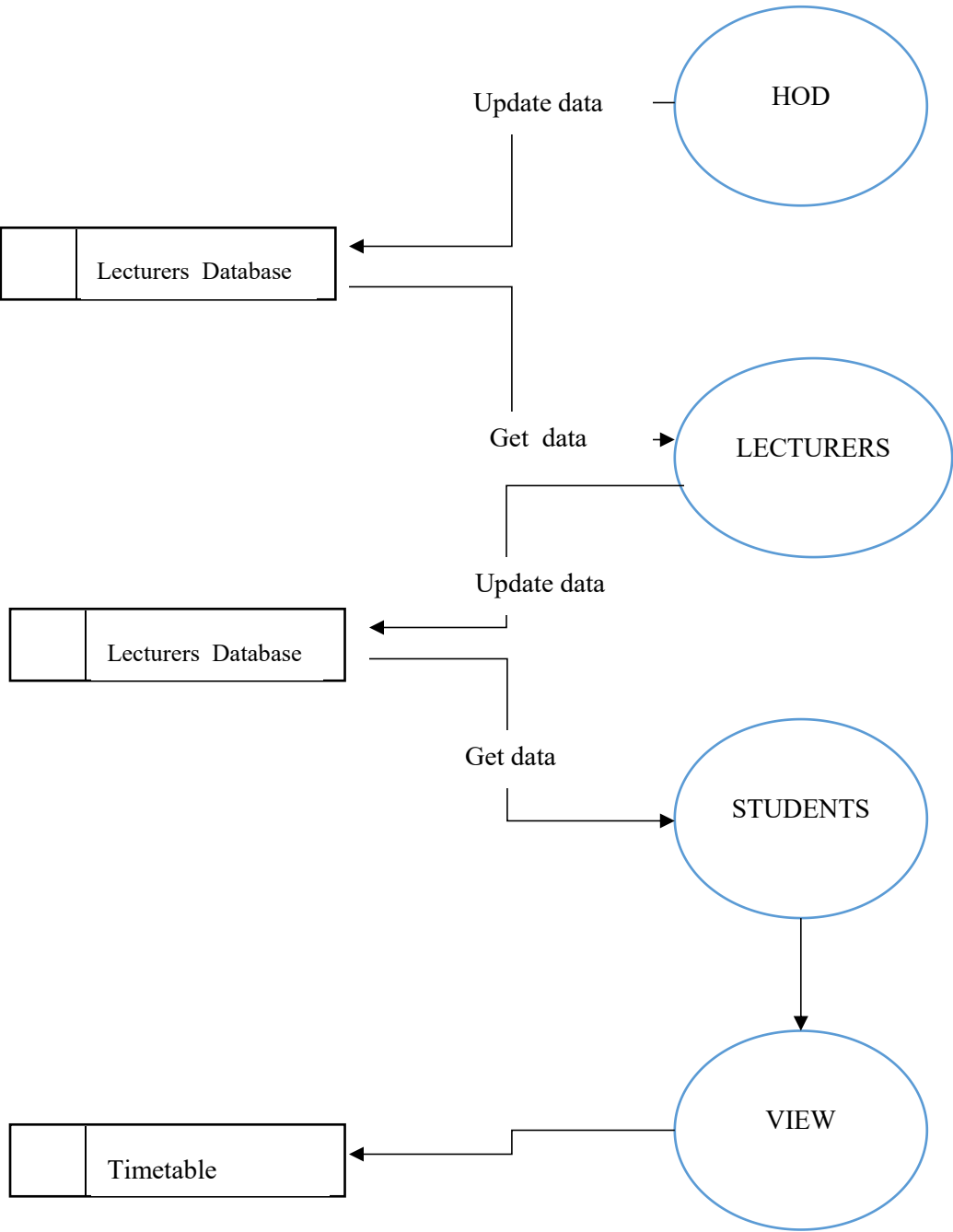


Figure 4:5: Dataflow Diagram



DATA FLOW DIAGRAM

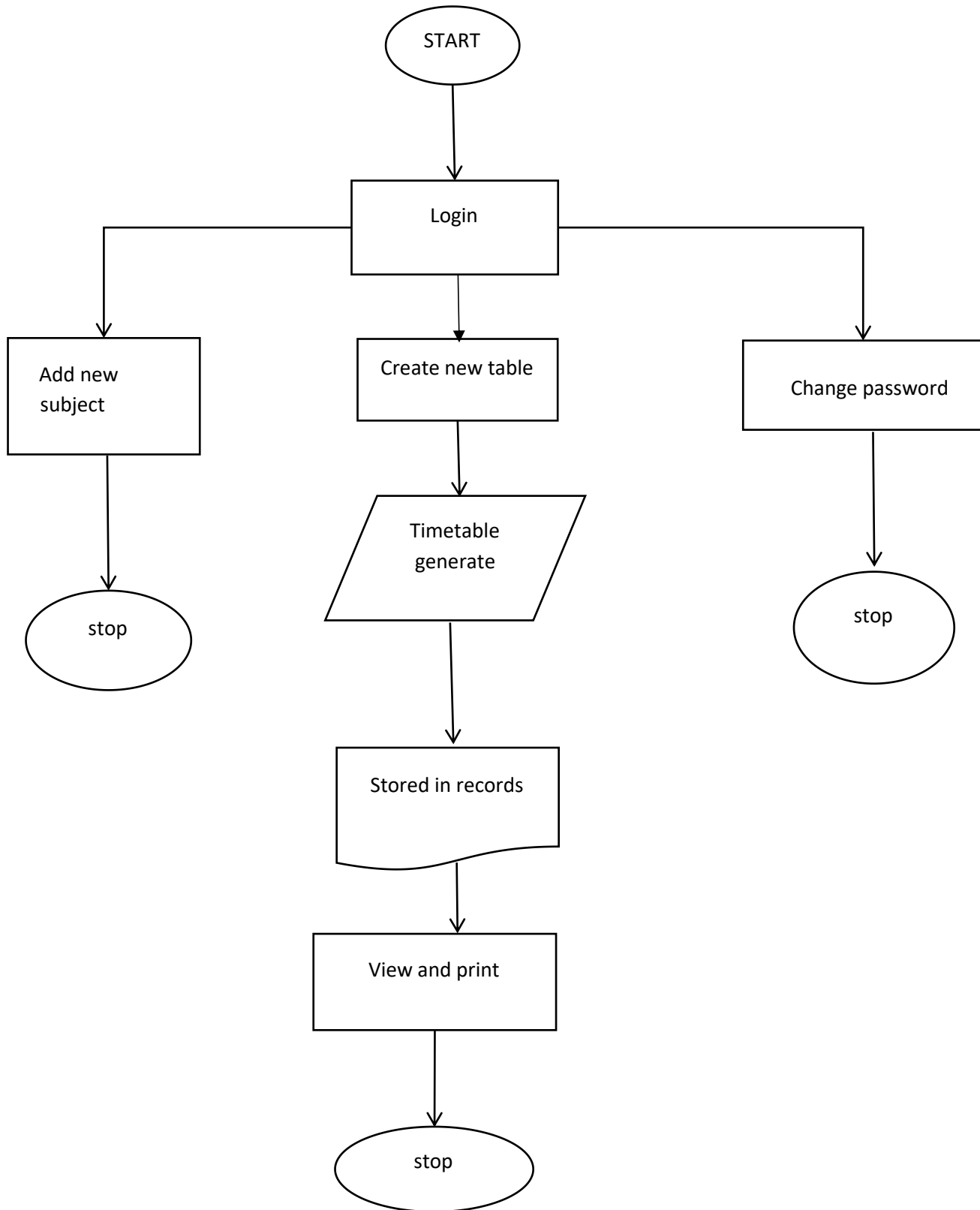


Figure 4:6: Dataflow Diagram : source GA@gmail.com.



1.24 Conclusion

In conclusion the researcher discussed how the proposed system is going to work and how it is going to be constructed or designed.



1.25 Coding

Introduction

The developer would introduce the programming language that would be used for development, programming style, connection method, processing method, input and output methods.

Programming language

The developer used Sublime text, MySQL connector, PHP, JavaScript and Hyper Text Mark-up Language for the development of the system.

```
<?php

    if(session_status() === PHP_SESSION_NONE){

        session_start();

    }

    $user = $_SESSION['user'];

    if($user == ""){

        header('location:login.php');

    }

    //link to menu nav bar

    include_once 'menu.html';

    //link to data base connection

    include_once 'atgs_connection.php';?>

<?php
```



```
try {
```

```
$tables = ['lecturers','timetabledata','subjects','departments','classes','timeslots'];
```

```
$rowCount = [];
```

```
$flag = 0;
```

```
foreach ($tables as $table) {
```

```
    $count_sql = "SELECT COUNT(*) AS row_count FROM $table";
```

```
    $exeRun = $con->query($count_sql);
```

```
    if($exeRun){
```

```
        $row = $exeRun->fetch_assoc();
```

```
        $rowCount[$flag] = "$row[row_count]";
```

```
    }
```

```
    else{
```

```
        echo"Error : ".$con->error;
```

```
    }
```

```
    $flag++;
```

```
}
```

```
} catch (Exception $e) {
```

```
    echo "<script>alert('Error: '$e)</script>";
```

```
}
```



?>

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <meta charset="utf-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
  <link rel="stylesheet" type="text/css" href="css/all.css">
```

```
  <link rel="stylesheet" type="text/css" href="atgs_styles.css">
```

```
  <title>ATGS Home</title>
```

```
</head>
```

```
<body>
```

```
  <form class="innerbody">
```

```
    <div class="outacontainer">
```

```
      <div class="innerindex">
```

```
        <div class="innerindexmessage">
```

```
          <h2><i class="fa-solid fa-chalkboard-teacher"></i>Lecturers</h2>
```

```
          <h1 id="lecturers"></h1>
```

```
        </div>
```

```
      <div class="innerindexmessage">
```

```
        <h2><i class="fa-solid fa-table"></i>Tables</h2>
```

```
        <h1 id="tables"></h1>
```

```
      </div>
```



Automated timetable generation

```
<div class="innerindexmessage">

    <h2><i class="fa-solid fa-chalkboard"></i>Subjects</h2>

    <h1 id="subjects"></h1>

</div>

<div class="innerindexmessage">

    <h2><i class="fa-solid fa-bars"></i>Departments</h2>

    <h1 id="dept"></h1>

</div>

<div class="innerindexmessage">

    <h2><i class="fa-solid fa-layer-group"></i>Classes</h2>

    <h1 id="course"></h1>

</div>

<div class="innerindexmessage">

    <h2><i class="fa-solid fa-clock"></i>TimeSlots</h2>

    <h1 id="slots"></h1>

</div>

</div>

</div>

</form>

<script type="text/javascript">

    //var tableNames = <?php echo json_encode($tables)?>;
```



Automated timetable generation

```
var tableValues = <?php echo json_encode($rowCount)?>;

document.getElementById('lecturers').innerHTML = addLeadingZero(tableValues[0]);

document.getElementById('dept').innerHTML = addLeadingZero(tableValues[3]);

document.getElementById('course').innerHTML = addLeadingZero(tableValues[4]);

document.getElementById('slots').innerHTML = addLeadingZero(tableValues[5]);

document.getElementById('tables').innerHTML = addLeadingZero(tableValues[1]);

document.getElementById('subjects').innerHTML = addLeadingZero(tableValues[2]);

function addLeadingZero(num){

    return new Intl.NumberFormat('en-US',{minimumIntegerDigits: 2}).format(num);

}

</script>

</body>

<!--*****ALL RIGHTS RESERVED © GEMRAT
TECHNOLOGIES*****-->

</html>
```

Testing

Before the system can be installed, implemented and integrated several testing processes have to be conducted in order to have bug and loop hole free system. Intensive test of the system was conducted to for instance, login and database data retrieval. Developer did some test to check errors which may affect the functionality and quality of the system.



Testing process

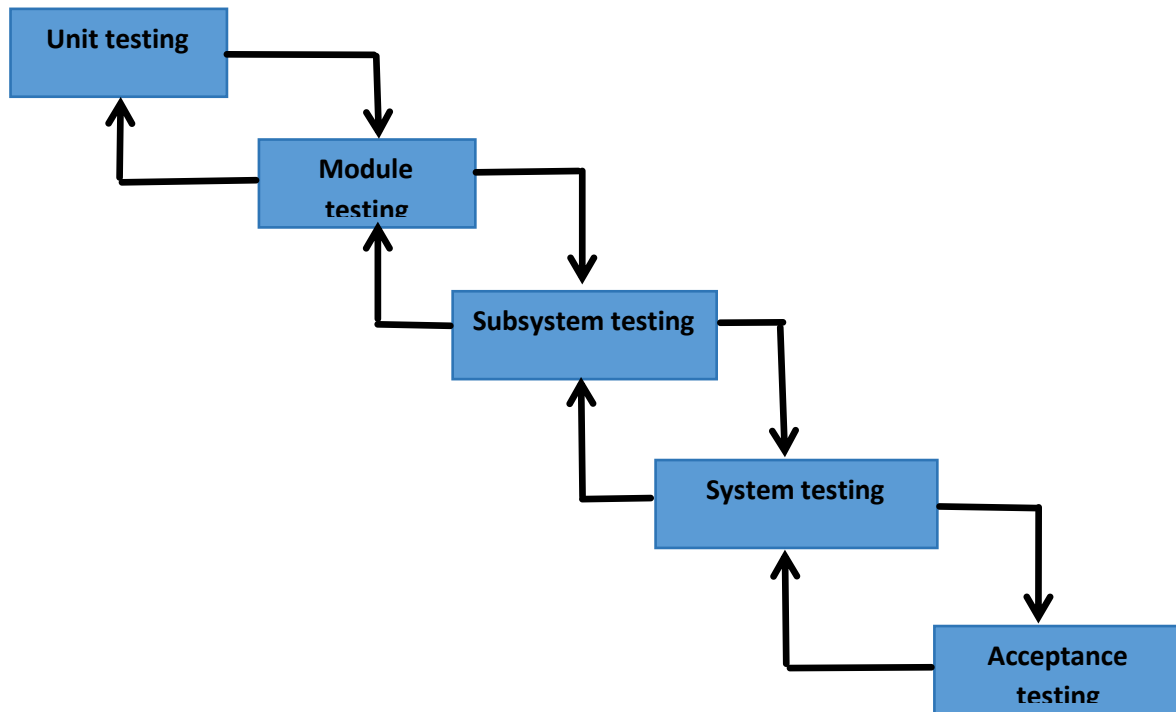


Figure 5:1: Testing Process

Module testing

This was also called link testing. It tests a collection of integrated modules into a subsystem. It ensures job streams are correct. There was a detection of interface mismatches and rigorous exercise of the interface between programs. The modules, which are linked together, were tested.

Subsystem testing

Subsystems were tested to see if they were performing well.

System testing

The system was tested as a whole and it worked fine.



Acceptance testing

User needed to taste system to assess whether a change and a simplified mode of message has been initiated to achieve goals. The system would be installed to various users not within the circular. In this stage errors and omissions that the developers might have missed out would be discovered.

Black box testing

The developer checked system functionality from input time process and output time. The system gave out the results as expected in the user requirements analysis.

White box testing

The developer identified the inner workings, structure, language of the system and its purpose.

Stress testing

The system was put under a lot of harsh conditions to see if it can sustain stress and pressure and everything worked perfect.

Interface testing

Interface was tested on different screen sizes and is fitted well on both PC and desktop computers. Validation was also tested as part of interface testing and everything worked perfect.

Acceptance testing

The developer tested the acceptance of the system through participation of the end user. To ensures that the system meets the functional requirements.



1.25.1 VALIDATION AND VERIFICATION

Validation

Developer ensured that the system meets the specified needs of the system. Data entered by the end users was validated to make it easier for them to use the system.

Verification

Developer evaluated whether the system complies with the requirement specification or not and ensured that the system functioned as required.

The screenshot displays the login interface for the 'ATGS KPP TIME TABLE GEN SYSTEM'. The title 'ATGS' is prominently shown in large white letters on a blue background. Below it, the text 'KPP TIME TABLE GEN SYSTEM(LOGIN)' is visible. A red error message, 'USERNAME OR PASSWORD INCORRECT', is centered on the screen. The login form consists of two yellow input fields: the first contains the username 'tsitsisau', and the second contains masked characters represented by dots. A 'LogIn' button is positioned at the bottom of the form.

Figure 5:2: Login verification



1.26 Implementation Post implementation plan

Introduction

The developer implemented the system and finding out whether the objectives of the users and the organization are met. Programs were coded using HTML, PHP, JavaScript and CSS, combined with MySQL.

Training

The developer provided a user manual for training users. Users were asked to operate the system in presence of system developer. This was done to monitor user's progress and comfort ability with the system. It was discovered that user was confident with system and was impressed with the functionality of it.

Training method

Well-detailed user manuals were provided to the system administrators and the users so that they would familiarize themselves with the operations of the system. In-house training was also done and the I.T department trained first such that they would provide technical support to the users who would need support after implementation.

User manual

The developer explained how to use the system in language that a non-technical person can understood as depicted by the steps bellow.



Login

ATGS

KPP TIME TABLE GEN SYSTEM(LOGIN)

ENTER USERNAME

ENTER PASSWORD

LogIn

Figure 6:1: Login

To log in into the system the user has to type tsitsisau as username and saunyama as password. After these correct entries, the user should click login button to be granted access to the main menu. If the user makes a typographical error, he/she can click the cancel button to start afresh.

Main menu

The main screen allowed access to all the subsystems of the Automated timetable system. The user chooses the option he /she wants to perform and works on the desired operation. The main menu has the following subsystems:



Automated timetable generation

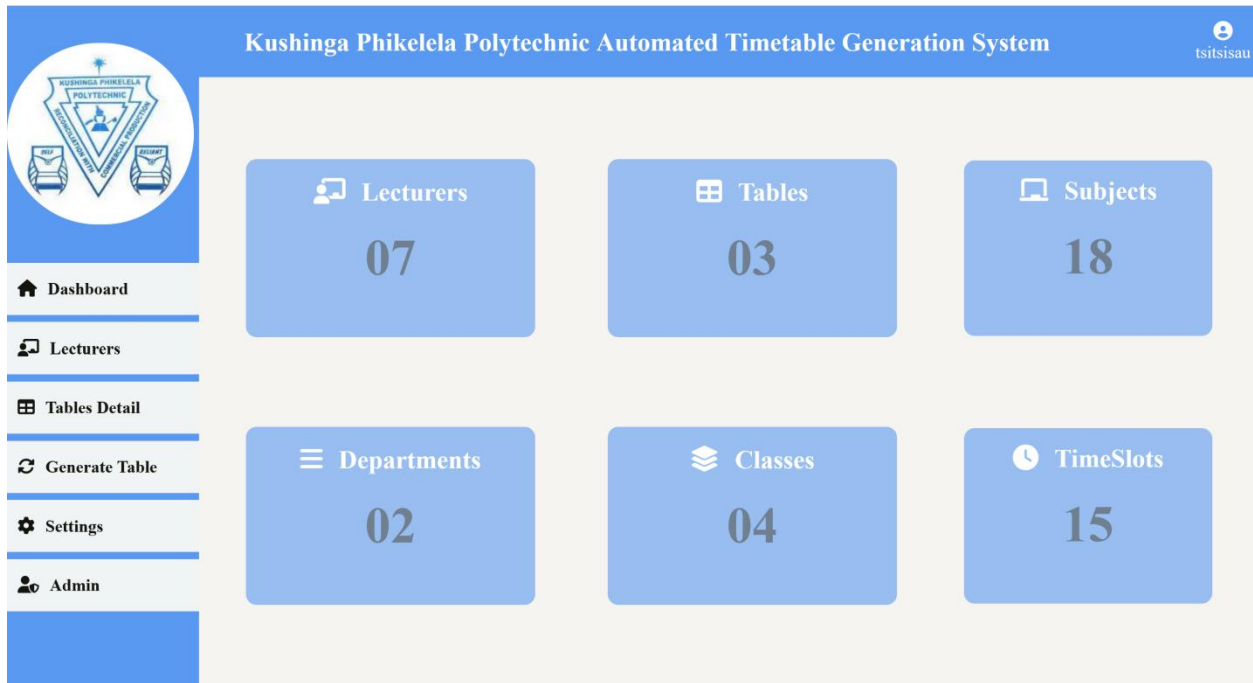


Figure 6:2 Main Menu

Conversion Plan

The developer described all activities that should occur during implementation of the new system and put it into operation.

Conversion methods

Parallel conversion

Old and new systems are used simultaneously. It provides fall-back when new system fails and offers greatest security and ultimately testing of new system.

Direct cut over conversion

New system to be implemented and the old system replaced completely. This strategy has relatively low cost of implementation, however there is an imminently high risk of the new system failing to meet the requirements or fail to give better functionalities than the old one.



Pilot Approach

Supported phased approach that gradually implement system across all users. It allows training and installation without unnecessary use of resources. Avoid large contingencies from risk management.

Phase-in method

Working version of system implemented in one part of organization based on feedback and installed throughout the organization all alone or stage by stage. It provides experience and line test before implementation when preferred new system involves new technology or drastically changes performance.

MAINTANANCE (POST IMPLEMENTATION)

The developer ensured that the newly implemented system meets the objectives for which it was developed to address. Periodic review, audit or monitoring of the system would be held with users so as to ensure that the system operates properly and meets the organization's objectives. The following are the types of maintenance.

Corrective Maintenance

The developer corrected parts of the system that would encounter some errors during the operation of the system. This would be done whenever there are reported errors into the system.

Perceptive Maintenance

The developer improved the system in order to maintain system effectiveness and efficient. The maintenance process would be triggered by the change request from the users and management.

Adaptive Maintenance

This would be done whenever some new environment has to be used. The system has to be changed sometimes when a different hardware platform or operating system has to be used.



System Review

The developer should correct errors in development or use of system. Monthly or yearly reviews should be held to address any problems arise.

System backup

The system would use hard drives, Compact Disks (CDs) and Flash memory for backing up information storage. These devices would be stored in a very secure place to protect them from physical damage.



1.27 FUTURE PLANS SUMMARY AND CONCLUSION

Future plans

Develop a mobile app that allow students and faculty to access their timetables and receive notifications about updates and changes . Tracks and manages lecturer's working hours per day and per week.Allow lecturers and students to give feedback about timetable feasibility. Develop a personalized timetabling system that takes into account individual student's preferences ,learning styles and goals.

1.28 SUMMARY

The review highlights that automated timetable generation systems have been successfully implemented in various educational institutions, with genetic algorithms and constraint-based models showing the most promise in handling complex scheduling tasks. These methods allow for the efficient allocation of resources, resolution of conflicts, and adaptability in the face of last-minute changes.

Challenges encountered

Gaining proficiency in the use of some of the programming tools: A lot of time was spent experimenting on programming language PHP and JavaScript as the system developer had little in-depth knowledge of the two. Limited application domain knowledge.



References

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- Ibrahim et al .2014 discussing challenges and solutions in automated timetable generation.
- Narang, Gupta and Bansal 2012 . genetic algorithms for optimizing timetable generation, covering theoretical models and implementations.
- Alhijawi and Kilani (2020). collaborative filtering recommender systems using genetic algorithms.
- Elsayed et al. (2014). solving optimization problems.
- Semenkin and Semenkina (2012). Worked on self – configuration genetic algorithms that introduced modified crossover operators to enhance performance.
- (Alkan & Ozcan, 2003) show that GA can reduce the likelihood of scheduling conflicts and increase resource utilization GA@gmail.com.
- Petrovic et al (2012) highlighted that timetable automation led to increased flexibility, allowing real-time changes in teaching schedules when unexpected events occurred.
- (Andersson & Flisberg, 2015) deployed a genetic algorithm-based timetable generator algorithm@gmail.com.



1.29 APPENDIX A

Questionnaire Students at Kushinga Phikelela

Date: 26/01/2024

The researcher sent the following questions to Students at Kushinga Phikelela

NB Please make use of ticks in the boxes where appropriate

1. Have you experienced clashes of timetable?

Yes ☐

No ☐

2. What methods are used to prevent the clashes of timetable?

.....

.....

.....

.....

3. Is data easily accessible and centralized?

Yes ☐

No ☐

4. Do you track lecturers, How?

.....

.....

.....

.....

5. Do you think timetable analysis is being done accurately and help full?

Yes ☐

No ☐



Interview questions for students at Kushinga Phikelela

Date: 21/01/2024

The researcher asked the following interview questions to students at Kushinga Phikelela:

1. How long does it take to have a timetable?

.....
.....
.....

Does timetable work ?

Satisfactory ☐ Excellent ☐ Poor ☐

1. Is the timetable experiencing rapid crashes or errors?

.....
.....
.....

2. What are the strengths and challenges of the current system?

.....
.....
.....



1.31 APPENDIX C

Observation

The researcher observed the following:

Observation guide schedule

Name of Observer: _____ Section:

Date: _29/07/2024_____ Time

Observation: _____

Conclusion:



1.32 APPENDIX D

Login Code

```
<?php
    if(session_status() === PHP_SESSION_NONE){
        session_start();
    }

    $user = $_SESSION['user'];
    if($user == ""){
        header('location:login.php');
    }
//link to menu nav bar
include_once 'menu.html';
//link to data base connection
include_once 'atgs_connection.php';?>
<?php
    try {

        $tables = ['lecturers','timetabledata','subjects','departments','classes','timeslots'];
        $rowCount = [];
        $flag = 0;

        foreach ($tables as $table) {
            $count_sql = "SELECT COUNT(*) AS row_count FROM $table";
            $exeRun = $con->query($count_sql);

            if($exeRun){
                $row = $exeRun->fetch_assoc();
                $rowCount[$flag] = "$row[row_count]";
            }
            else{
                echo "Error : ".$con->error;
            }
            $flag++;
        }
    } catch (Exception $e) {
        echo "<script>alert('Error: '$e)</script>";
    }
?>
<!DOCTYPE html>
```



```
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet" type="text/css" href="css/all.css">
  <link rel="stylesheet" type="text/css" href="atgs_styles.css">
  <title>ATGS Home</title>
</head>
<body>
  <form class="innerbody">
    <div class="outacontainer">
      <div class="innerindex">
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-chalkboard-teacher"></i>Lecturers</h2>
          <h1 id="lecturers"></h1>
        </div>
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-table"></i>Tables</h2>
          <h1 id="tables"></h1>
        </div>
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-chalkboard"></i>Subjects</h2>
          <h1 id="subjects"></h1>
        </div>
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-bars"></i>Departments</h2>
          <h1 id="dept"></h1>
        </div>
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-layer-group"></i>Classes</h2>
          <h1 id="course"></h1>
        </div>
        <div class="innerindexmessage">
          <h2><i class="fa-solid fa-clock"></i>TimeSlots</h2>
          <h1 id="slots"></h1>
        </div>
      </div>
    </div>
  </form>
  <script type="text/javascript">

    //var tableNames = <?php echo json_encode($tables)?>;
```



Automated timetable generation

```
var tableValues = <?php echo json_encode($rowCount)?>;
```

```
document.getElementById('lecturers').innerHTML = addLeadingZero(tableValues[0]);  
document.getElementById('dept').innerHTML = addLeadingZero(tableValues[3]);  
document.getElementById('course').innerHTML = addLeadingZero(tableValues[4]);  
document.getElementById('slots').innerHTML = addLeadingZero(tableValues[5]);  
document.getElementById('tables').innerHTML = addLeadingZero(tableValues[1]);  
document.getElementById('subjects').innerHTML = addLeadingZero(tableValues[2]);
```

```
function addLeadingZero(num){  
    return new Intl.NumberFormat('en-US', {minimumIntegerDigits: 2}).format(num);  
}
```

```
</script>
```

```
</body>
```

```
<!--*****ALL RIGHTS RESERVED © GEMRAT  
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```

```
</html>
```